

E.1 / PAPER 1 / QUESTION 16

29 April 2025 / TZ2 (file label) / HL Paper 1A / original Q35 / 2225-6713

- 16.** The intensity of electromagnetic radiation incident on a surface is I . The wavelength of the radiation is doubled and the number of photons incident per unit time per unit area is doubled.

What is the new intensity of radiation incident on the surface?

- A. $\frac{I}{2}$
- B. I
- C. $2I$
- D. $4I$